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Let's Think Dot by Dot: Hidden Computation in Transformer Language Models

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Abstract

Chain-of-thought responses from language models improve performance across most benchmarks. However, it remains unclear to what extent these performance gains can be attributed to human-like task decomposition or simply the greater computation that additional tokens allow. We show that transformers can use meaningless filler tokens (e.g., '.....') in place of a chain of thought to solve two hard algorithmic tasks they could not solve when responding without intermediate tokens. However, we find empirically that *learning* to use filler tokens is difficult and requires specific, dense supervision to converge. We also provide a theoretical characterization of the class of problems where filler tokens are useful in terms of the *quantifier depth* of a first-order formula. For problems satisfying this characterization, chain-of-thought tokens need not provide information about the intermediate computational steps involved in multi-token computations. In summary, our results show that additional tokens can provide computational benefits independent of token choice. The fact that intermediate tokens can act as filler tokens raises concerns about large language models engaging in unauditable, hidden computations that are increasingly detached from the observed chain-of-thought tokens.[1]

1 Introduction

Chain-of-thought reasoning improves language model (LM) performance when compared to direct, no chain-of-thought, responses (Wei et al., 2023; Suzgun et al., 2022; Lanham et al., 2023). However, recent empirical work shows that answers arrived at via chains of thought frequently are not faithful to the intermediate reasoning steps taken within the chain (Lanham et al., 2023; Turpin et al., 2023). As a limit case of unfaithfulness, the *filler token* setting replaces chain-of-thought tokens with arbitrary, repeated tokens, e.g. '.....', as shown in Figure 1. By comparing language model performance when given filler tokens instead of chains of thought, we can assess whether a given LM is capable of carrying out cross-token computations that are not reflected in the chain of thought tokens.

The most widely used LM alignment methods are purely behavioral. Reinforcement learning from human feedback, constitutional AI, instruction fine-tuning, and automated red-teaming all rely on judging or comparing model output tokens. LMs capable of making use of filler tokens undermine this reliance because the reasoning carried out across filler tokens cannot be judged from the tokens themselves.

In this work, we study the strict filler case where filler tokens are repeated dots, '.....'; however, the utility of such tokens depends only on the availability of excess capacity in activation space. The '.....' case is a minimal version of the more general setting where any sequence of filler tokens is provided between an input prompt and some complex output token. For example, the filler sequence could be "Lorem ipsum dolor sit amet, ..." or

[1] Code is available at https://github.com/JacobPfau/fillerTokens

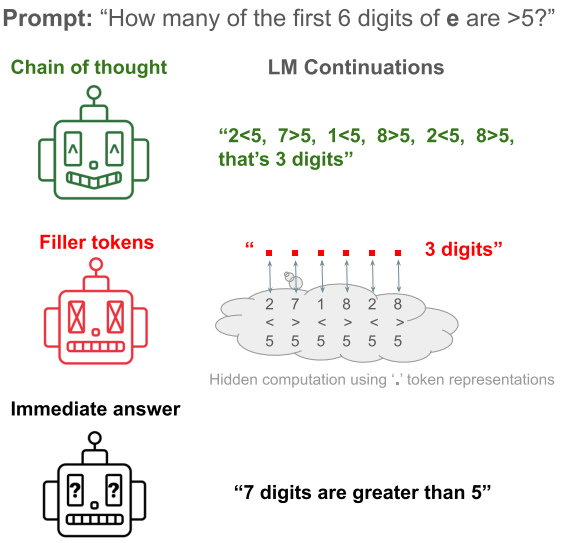

Figure 1: A stylized example distinguishing between three LM question-answering protocols: chain of thought, filler tokens, and immediate answer. In the filler-tokens setting, the LM uses arbitrary, irrelevant intermediate tokens (e.g., '.....') before answering, but the filler tokens' hidden-layer representations still provide computation relevant to later tokens. In previous work, chain-of-thought was shown to allow greater expressive power than immediate answer (correspondingly, the 'immediate answer' bot gives the wrong answer). We show that filler tokens can, on certain tasks, match the performance of chain-of-thought reasoning.

repeating a question back to the user, as long as the string requires minimal computation and precedes a more algorithmically demanding token.

Empirically, commercial large language models (LLMs) do not benefit from filler tokens on common QA and math benchmarks; Claude 2 and GPT-3.5 achieve the same performance with filler tokens as they do when responding directly without intermediate tokens (Sachan, 2023; Lanham et al., 2023). However, current LLMs' limitations cannot be extrapolated to larger scales: The empirical evidence on current LLMs does not clarify whether a failure to use filler tokens is an in-principle limitation of transformer expressivity (or their loss landscapes), or instead, if filler token performance may arise at larger scale. Additionally, it is unclear whether these evaluations targeted tasks where filler tokens would be beneficial. In this work, we demonstrate that transformers trained on the next-token prediction objective *can achieve improved performance on certain tasks when given filler tokens*, achieving perfect accuracy whereas the no-filler, immediate-answer setting achieves only low accuracy.

These results also provide interesting insight into how filler tokens extend the expressive power of transformers. As single-token predictors, transformers can only solve problems in a complexity class called TC^0, which means transformers cannot express problems like permutation composition or graph connectivity (Merrill & Sabharwal, 2023a; Strobl et al., 2023). Whereas linear or polynomial chain-of-thought steps can add power to transformers beyond TC^0 (Merrill & Sabharwal, 2023a), transformers remain in TC^0 with even a polynomial number of filler tokens. Thus, unlike for chain of thought, we cannot expect filler tokens to let transformers solve problems outside TC^0, e.g. graph connectivity. However, our results suggest that *filler tokens likely extend the expressive power of transformers within* TC^0. In particular, our results establish that reasoning requiring many *nested quantifiers* becomes expressible for transformers with filler tokens whereas it is conjectured that no-intermediate-token, immediate-answer transformers cannot solve these problems. We propose synthetic tasks for which transformers without chain of thought have been conjectured inadequate in

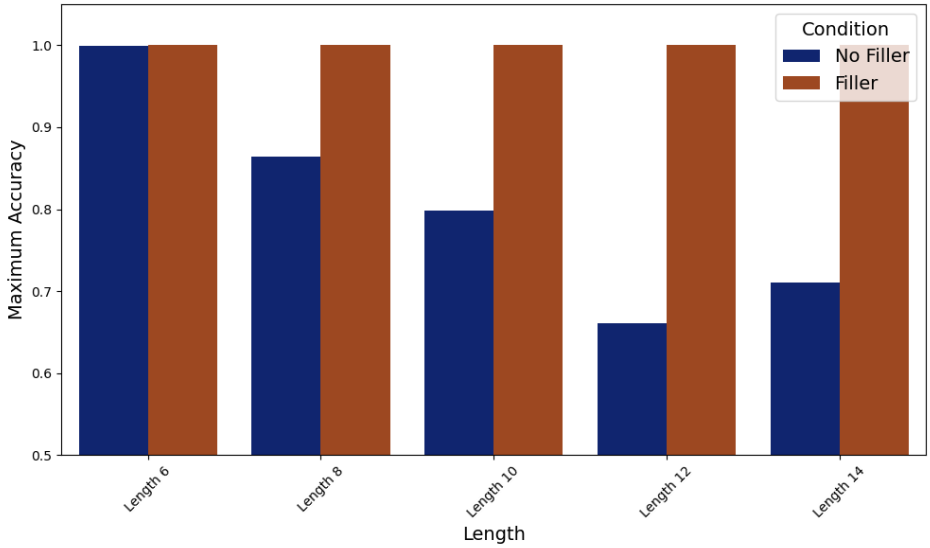

Figure 2: The performance gap between a transformer, Llama 34M, with and without filler tokens increases with 3SUM problem length up to length 12, showing that filler tokens reliably provide an advantage for sufficiently complex 3SUM problems. The no-filler models were trained for $5\times$ the number of steps.

expressivity (Sanford et al., 2024) and show that *using filler tokens, transformers can solve these tasks.*

Our contributions are the following:

1. We construct two synthetic datasets, 3SUM (Figure 3) and 2SUM-Transform, on which LLAMA transformers fail to solve the task without filler, but achieve 100% and 94% accuracy, respectively, when provided filler tokens.
2. We find that filler token performance increases over immediate answers as the length and complexity of inputs increase (Figures 2 and 5).
3. We contextualize filler tokens with respect to theoretical expressivity results highlighting that filler-token prompting remains within circuit complexity class TC^0, but we show empirically that they do seem to add power *within* TC^0.
4. We find that learning to use filler tokens is difficult and requires specific, dense supervision to converge. Standard chain-of-thought data is insufficient for models to learn to leverage filler tokens effectively, c.f. Section 4.3.

Taken together these findings suggest that although current LLMs are unlikely to benefit from filler tokens, this is not an in-principle limitation of current architectures. Given demonstrations of parallelizable task decompositions, we expect that current LLMs would also realize benefits from filler tokens.

2 Related Work

Transformer Expressivity and Filler Tokens Recent theoretical work establishes that transformers without additional reasoning tokens are limited to solving only highly parallelizable problems (see Strobl et al., 2023 for an overview). Formally, Merrill & Sabharwal (2023a) place log-precision transformers in the circuit complexity class TC^0, which can be equivalently understood as the class of problems definable in first-order logic with majority quantifiers (Merrill & Sabharwal, 2023b). It follows that problems outside TC^0 (those that cannot be defined in first-order majority logic) *cannot* be solved by transformers